Q1: 16 March (Shift 2) - Single Correct

The half-life of Au^{198} is 2.7 days. The activity of 1.50mg of Au^{198} if its atomic weight is 198 g mol^{-1} is, $\left(N_A = 6 \times 10^{23}/mol\right)$

(1) 240Ci

(2) 357Ci

(3) 535Ci

(4) 252Ci

Q2: 16 March (Shift 2) - Single Correct

Calculate the time interval between 33% decay and 67% decay if half-life of a substance is 20 minutes.

(1) 60 minutes

(2) 20 minutes

(3) 40 minutes

(4) 13 minutes

Q3: 18 March (Shift 1) - Single Correct

A radioactive sample disintegrates via two independent decay processes having half lives $T_{1/2}^{(1)}$ and $T_{1/2}^{(2)}$ respectively. The effective half- life $T_{1/2}$ of the nuclei is:

(1) None of the above

(2) $T_{1/2} = T_{1/2}^{(1)} + T_{1/2}^{(2)}$

(3) $T_{1/2} = \frac{T_{1/2}^{(1)} T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$

(4) $T_{1/2} = \frac{T_{1/2}^{(t)} + T_{1/2}^{(2)}}{T_{1/2}^{(1)} - T_{1/2}^{(2)}}$

Q4: 18 March (Shift 2) - Single Correct

The decay of a proton to neutron is :

(1) not possible as proton mass is less than the neutron mass

(2) possible only inside the nucleus

(3) not possible but neutron to proton conversion is possible

(4) always possible as it is associated only with β^+ decay

Answer Key

Q1 (2) **Q2** (2) **Q3** (3) **Q4** (2)

Q1 (2)

$A = \lambda N$

$N = nN_A$

$N = \left(\frac{1.5 \times 10^{-3}}{198}\right) N_A$

$A = \left(\frac{\ln 2}{t_{1/2}}\right) N$

$1 \text{ Curie} = 3.7 \times 10^{10} \text{ Bq}$

$A = 365 \text{ Bq}$

Q2 (2)

$N_1 = N_0 e^{-\lambda t_1}$

$\frac{N_1}{N_0} = e^{-\lambda t_1}$

$0.67 = e^{-\lambda t_1}$

$\ln(0.67) = -\lambda t_1$

$N_2 = N_0 e^{-\lambda t_2}$

$\frac{N_2}{N_0} = e^{-\lambda t_2}$

$0.33 = e^{-\lambda t_2}$

$\ln(0.33) = -\lambda t_2$

$\ln(0.67) - \ln(0.33) = \lambda t_1 - \lambda t_2$

$\lambda (t_1 - t_2) = \ln\left(\frac{0.67}{0.33}\right)$

$\lambda (t_1 - t_2) \cong \ln 2$

$t_1 - t_2 \simeq \frac{\ln 2}{\lambda} = t_{1/2}$

Half life $= t_{1/2} = 20$ minutes.

Q3 (3)

$\lambda_{eq} = \lambda_1 + \lambda_2$

$\frac{1}{T_{1/2}} = \frac{1}{T_{1/2}^{(1)}} + \frac{1}{T_{1/2}^{(2)}}$

$T_{1/2} = \frac{T_{1/2}^{(1)} T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$

Q4 (2)

It is possible only inside the nucleus and not otherwise.